



iStocks

AI-Powered Stock Analysis Platform

Technical Documentation & Feature Overview

Version 1.0

A comprehensive intelligent stock analysis platform
combining real-time market data with AI-driven insights

January 2026

Live at: <https://istocks.codes>

Contents

1	Executive Summary	2
2	Platform Features	2
2.1	AI-Powered Chatbot	2
2.2	Real-Time Stock Dashboard	3
2.3	Technical Indicators	3
2.4	Sangraha - Strategy Repository	3
2.5	User Authentication	3
2.6	Database Chat	4
3	Technical Architecture	4
3.1	System Overview	4
3.2	Technology Stack	4
3.3	Database Schema	5
4	API Endpoints	5
5	Security Implementation	6
6	Future Roadmap & Suggestions	6
6.1	Short-Term Enhancements	6
6.2	Medium-Term Features	6
6.3	Long-Term Vision	6
6.4	Infrastructure Recommendations	7
7	Deployment & DevOps	7
8	Conclusion	7

1 Executive Summary

iStocks is a cutting-edge financial technology platform designed to democratize institutional-grade stock analysis for retail investors in the Indian market. The platform combines real-time market data from NSE/BSE with advanced AI capabilities powered by Google's Gemini model to deliver actionable investment insights.

Key Value Propositions

- **AI-Driven Analysis:** Natural language queries to understand complex market dynamics
- **Real-Time Data:** Live stock prices and technical indicators
- **Strategy Repository:** Create, backtest, and share trading strategies
- **Technical Analysis:** 20+ indicators including RSI, MACD, Bollinger Bands
- **Zero Cost Barrier:** Free access to professional-grade tools

2 Platform Features

2.1 AI-Powered Chatbot

The heart of iStocks is its intelligent conversational AI assistant that understands financial queries in natural language.

Capabilities:

- **Natural Language Processing:** Ask questions like “How is Reliance performing?” or “Compare TCS vs Infosys”
- **Technical Analysis Interpretation:** Explains RSI, MACD, and other indicators in simple terms
- **Pattern Recognition:** Identifies bullish/bearish patterns and trends
- **Multi-Stock Comparison:** Compare multiple stocks simultaneously
- **Volume Analysis:** Analyzes trading volumes and identifies unusual activity

AI Model: Google Gemini 2.5 Pro - chosen for its superior reasoning capabilities and context understanding.

2.2 Real-Time Stock Dashboard

Dashboard Components	
Price Card	Live price, change percentage, high/low with color-coded indicators
Interactive Chart	Candlestick and line charts with zoom, pan capabilities
Technical Panel	20+ indicators with visual representations
Insights Panel	AI-generated analysis and recommendations

2.3 Technical Indicators

The platform calculates and displays comprehensive technical indicators:

Category	Indicators
Moving Averages	SMA (20, 50, 200), EMA (12, 26)
Momentum	RSI, Stochastic Oscillator, Williams %R, ROC
Trend	MACD, ADX, Plus/Minus DI
Volatility	Bollinger Bands (Upper, Middle, Lower), ATR
Volume	OBV (On-Balance Volume), Volume Analysis
Others	CCI (Commodity Channel Index)

Table 1: Technical Indicators Available on iStocks

2.4 Sangraha - Strategy Repository

“Sangraha” (Sanskrit for “collection”) is a unique feature allowing users to create, share, and backtest trading strategies.

Core Features:

- **Strategy Builder:** Create custom strategies with entry/exit rules
- **Backtesting Engine:** Test strategies against historical data
- **Performance Metrics:** Win rate, profit factor, max drawdown, Sharpe ratio
- **Community Sharing:** Fork and modify public strategies
- **Version Control:** Track strategy modifications over time

2.5 User Authentication

Secure authentication system with multiple options:

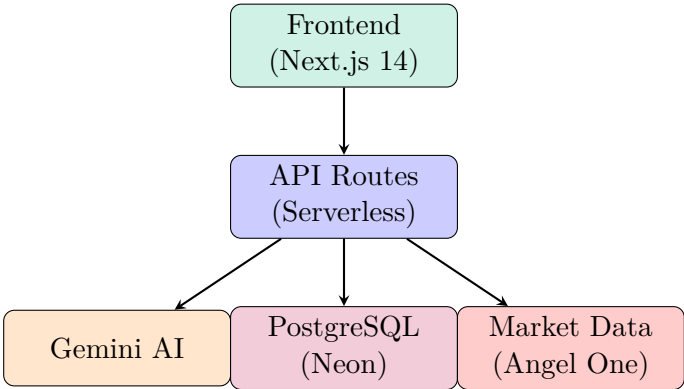
- Google OAuth 2.0 integration
- Email/Password with secure bcrypt hashing
- Session management with NextAuth.js
- Auto-verification for seamless onboarding

2.6 Database Chat

Advanced feature for power users to query stock data using natural language, which gets translated to SQL queries for precise data retrieval.

3 Technical Architecture

3.1 System Overview



3.2 Technology Stack

Frontend Technologies	
Framework	Next.js 14 with App Router
Language	TypeScript (strict mode)
Styling	Tailwind CSS with custom design system
Charts	Recharts for interactive visualizations
Icons	Lucide React icon library
State	React hooks with server components

Backend Technologies	
Runtime	Node.js with TypeScript
API	Next.js API Routes (Serverless)
ORM	Prisma with type-safe queries
Authentication	NextAuth.js v4
Password Hashing	bcryptjs
Email	Nodemailer with Gmail SMTP

Database & Infrastructure	
Database	PostgreSQL (Neon - Serverless)
Hosting	Vercel (Edge Network)
Domain	istocks.codes (Custom SSL)
Cron Jobs	Vercel Cron for scheduled updates
Version Control	Git with GitHub

AI & Data Processing	
AI Model	Google Gemini 2.5 Pro
AI SDK	Vercel AI SDK
Market Data	Angel One API
Technical Analysis	technicalindicators library
Data Scripts	Python for bulk processing

3.3 Database Schema

Key entities in the database:

- **Stock:** Symbol, name, exchange, price data
- **StockPrice:** OHLCV data with 20+ technical indicators
- **User:** Authentication and profile data
- **Account/Session:** OAuth and session management
- **Strategy:** User-created trading strategies
- **Backtest:** Strategy backtesting results
- **ChatSession/ChatMessage:** Conversational history

4 API Endpoints

Method	Endpoint	Description
POST	/api/chat	AI chatbot interactions
GET	/api/stocks	List all available stocks
GET	/api/stocks/[symbol]	Get stock details and prices
POST	/api/stocks/[symbol]/analyze	AI analysis for specific stock
GET	/api/sangraha	List all strategies
POST	/api/sangraha	Create new strategy
POST	/api/sangraha/[id]/backtest	Run backtest on strategy
POST	/api/sangraha/[id]/fork	Fork a public strategy
POST	/api/auth/signup	User registration
GET/POST	/api/auth/[...nextauth]	NextAuth handlers
GET	/api/cron/update-stocks	Scheduled stock data update

Table 2: Core API Endpoints

5 Security Implementation

Security Measures

- **Password Security:** bcrypt hashing with salt rounds
- **OAuth 2.0:** Secure Google authentication flow
- **Session Management:** Secure JWT tokens with NextAuth
- **HTTPS:** SSL/TLS encryption on all endpoints
- **Input Validation:** Zod schema validation on all inputs
- **SQL Injection Prevention:** Prisma's parameterized queries
- **Environment Variables:** Secrets stored in Vercel

6 Future Roadmap & Suggestions

6.1 Short-Term Enhancements

1. **Portfolio Tracking:** Allow users to create and track virtual portfolios
2. **Price Alerts:** Push notifications for price movements
3. **Watchlists:** Personal stock watchlists with quick access
4. **Market News Integration:** Real-time financial news feed
5. **Export Reports:** PDF/Excel export of analysis

6.2 Medium-Term Features

1. **Paper Trading:** Simulated trading with virtual money
2. **Strategy Marketplace:** Paid premium strategies
3. **Social Features:** Follow traders, share insights
4. **Options Analysis:** Options chain and greeks calculator
5. **Multi-Exchange Support:** Add MCX, commodity markets

6.3 Long-Term Vision

1. **Mobile Application:** Native iOS/Android apps
2. **Broker Integration:** Direct trading via broker APIs
3. **ML-Based Predictions:** Custom trained models for price prediction
4. **Voice Assistant:** Voice-based stock queries
5. **API for Developers:** Public API for third-party apps

6.4 Infrastructure Recommendations

- **AWS Migration:** Move compute-intensive backtesting to AWS Lambda/EC2
- **Redis Cache:** Add caching layer for frequently accessed data
- **CDN:** Utilize AWS CloudFront for static assets
- **Message Queue:** SQS for async processing of large backtests
- **Monitoring:** Implement AWS CloudWatch for observability

7 Deployment & DevOps

Current Infrastructure	
Platform	Vercel (Hobby Plan)
Database	Neon PostgreSQL (Serverless)
Domain	istocks.codes (Cloudflare DNS)
CI/CD	GitHub Actions + Vercel Auto-Deploy
Monitoring	Vercel Analytics

Deployment Flow:

1. Push code to GitHub main branch
2. Vercel automatically triggers build
3. Prisma generates client during build
4. Next.js compiles and optimizes
5. Deploy to global edge network

8 Conclusion

iStocks represents a significant step forward in democratizing financial analysis for Indian retail investors. By combining the power of AI with real-time market data and an intuitive interface, the platform enables users to make more informed investment decisions.

The modular architecture and modern tech stack ensure scalability and maintainability as the platform grows. With the planned roadmap items, iStocks is positioned to become a comprehensive financial intelligence platform.



iStocks

Empowering Investors with AI

Live: <https://istocks.codes>

GitHub: <https://github.com/i24hour/istocks-p>

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